

Inputs

AIS stream
(106/min)

OSM tiles
(coast, pier)

Anchor batch
(lat, lon, t)

M-CTX core (3 indices)

BR-LZ OSM Range
(Sec V)

Linear-time SDF
(Sec VI)

B^x -tree kNN
(Sec VII)

JCX
(Appendix A)
optional

Per-anchor context

ids_{shore}
(STR-tree)

SDF tensor
 128×128 f16

neighbours
+recall=1

Drop-in

Downstream
LSTM/Tx

Joint sketch

